

Alexander Bowring

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Senior Mathematical Consultant and Oxford PhD with 9 years experience in advanced statistics, ML, and optimisation. Technical lead on **£2M+** projects for national energy, FMCG and utility enterprises, with deployed models in the UK electricity control room saving **£1.7M** per month in operational costs.

Professional Experience

Senior Mathematical Consultant (previously Mathematical Consultant)

Oxford, UK

Smith Institute

Nov 2021 – Present

Led cross-functional teams (**6-10** members) across a **£2M+** project portfolio. Key projects include:

- **Dynamic Reserve Setting** (client: NESO): Built end-to-end explainable gradient-boosted ML models to forecast national energy reserves, engaging with stakeholders from engineers to executives through all iterations.
 - Deployed in UK national electricity control room, reducing reserve procurement by **300MW** every **30** minutes and saving **£1.7M** per month.
 - Awarded [Operational Research Society President's Medal 2024](#) for Best Application of Operational Research.
- **Vulnerability Future Energy Scenarios** (client: SSEN): Built gradient boosted ML models, applied SHAP explainability and advanced clustering to identify common drivers of vulnerability in communities served by SSEN.
 - Shaped SSEN's network investment strategy, informing decisions on low-carbon transition and targeted support of vulnerable communities.
 - Awarded [Utility Week Unlocking Data Award 2024](#) and [DataIQ Award 2024](#) for Best Use of AI for the Public Good.
- **Dispatch Decision Intelligence** (client: NESO): Delivered performance, functional and explainability upgrades to optimisers used in the national electricity control room for procuring generation to meet national demand.
 - Enhanced optimiser performance, situational awareness, and decision confidence through a novel explainability interface for querying optimiser decisions. Aspects of solution deployed in **3** national control room optimisers.
- **Weather Driven Sales Forecasting** (client: Coca-Cola European Partners): Parallelised model pipelines and led overhaul of data ingestion system, reducing runtime by **18%** and ensuring critical sales insights delivery.
 - Weekly reports actively used to guide CCEP's sales and distribution strategy across **9** European national markets.

Research Fellow in Neuroimaging Statistics

Oxford, UK

Nuffield Department of Population Health, University of Oxford

Nov 2019 - Nov 2021

Conducted reproducibility research on fMRI analysis variability across **59** pipelines using Python, Matlab, and Unix.

Published in *Nature* and *Human Brain Mapping*.

Education

Doctor of Philosophy in Population Health

Oxford, UK

Nuffield Department of Population Health, University of Oxford

Oct 2016 - Nov 2019

Developed a novel statistical method to identify neuronal activations in fMRI data, and analysis results reproducibility across analysis software. Published in *NeuroImage* and *Human Brain Mapping*, gave oral presentations at international OHBM conference (<**5%** of abstracts selected for oral presentation) and awarded **£2,000** Merit Abstract Award.

Bachelor of Science in Mathematics

University of Warwick, UK

University of Warwick

Sep 2012 - Jul 2015

Graduated with **First Class Honours**, achieving >**80%** in Algebra II, Analysis III, Galois Theory, Programming for Scientists.

Skills

Machine Learning & Statistics: Gradient boosted trees (LightGBM, XGBoost), SHAP explainability, clustering, Optuna hyperparameter tuning, reproducibility methods, time-series modelling and experimental design

Optimisation: Linear and mixed-integer programming, Gurobi, explainable optimisation solutions

Programming & Data Analysis: Python (pandas, numpy, sklearn, Jupyter, PyTorch), R (tidyverse, officer), SQL, bash

Cloud & DevOps: AWS (EC2, S3, CodeCommit), Azure DevOps, MLOps, CI/CD pipelines, Git/GitHub

Data Visualisation & BI: Power BI, Neo4j, matplotlib, ggplot2, interactive dashboards, report automation

Publications

I have published academic research in several high-impact journals including *Nature*, *NeuroImage*, and *Human Brain Mapping*. Full list available at [Google Scholar](#).